

Polynomial



terms

Many

i) $4x^3, 7xy, 9, -5xyz$ = Monomial

ii) $4x + y$ = Binomial

iii) $x + y + z$ = Trinomial

∴ Quadratic Equation

$$ax^2 + bx + c = 0$$

[a, b, c = real no.
a ≠ 0]

① 2 roots

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

(i) $\alpha, \beta = \text{roots}$

(ii) Sum of roots = $\alpha + \beta = -b/a$

(iii) Product of roots = $\alpha\beta = c/a$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\alpha = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$$

$$\beta = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

$$\Rightarrow \alpha + \beta = \frac{-\cancel{2}b}{\cancel{2}a} = -\frac{b}{a}$$

$$\Rightarrow \alpha \cdot \beta = \frac{b^2 - (b^2 - 4ac)}{4a^2} = \frac{\cancel{4}ac}{\cancel{4}a^2} = \frac{c}{a} //$$

$$\rightarrow x^2 - 5x + 6 = 0$$

$\rightarrow$ Roots : ?

i) $x = a \rightarrow \text{LHS} = \text{RHS}$

ii) $f(x) = 0$

then $x = a \rightarrow f$ is zero

$$\rightarrow \text{factors} : (x - 3)(x - 2) = 0$$

$\rightarrow$ Roots :

$$x = 2, 3$$

$$\cdot \# \quad ax^2 + bx + c = 0$$

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0$$

$$x^2 - \left[-\frac{b}{a}\right]x + \left[\frac{c}{a}\right] = 0$$

$$x^2 - [\text{Sum of roots}]x + [\text{Product of roots}] = 0$$

$$ax^2 + bx + c = 0$$

Nature of roots



Discriminant



$$D = b^2 - 4ac$$

$$\left[\begin{array}{l} = 0 \\ > 0 \\ < 0 \end{array} \right.$$

Real & equal

Real & distinct

Imaginary &
conjugate pair

Roots

Q] For what values of k will $4x^5 + 9x^4 - 7x^3 - 5x^2 - 4kx + 3k^2$ contain $(x-1)$ as a factor?

① $(x-1)$ is factor $\rightarrow x-1=0 \rightarrow x=1$ (Root)

② $f(x) = 4x^5 + 9x^4 - 7x^3 - 5x^2 - 4kx + 3k^2$

$$\boxed{f(1) = 0}$$

$$4 + 9 - 7 - 5 - 4k + 3k^2 = 0$$

$$3k^2 - 4k + 1 = 0$$

$$\rightarrow k = \frac{-(-4) \pm \sqrt{16 + 12}}{2(3)}$$

$$= 1, \frac{1}{3} //$$

Q] If the roots of the equation $Ax^2 + Bx + C = 0$ are -1 and 1 then which one of the following is correct?

- (A) A and C are both zero
- ~~(B) A and B are both positive~~
- (C) A and C are both negative
- ~~(D) A and C are of opposite sign~~

Solⁿ:

$$\textcircled{1} \quad \alpha + \beta = -b/a = -B/A$$

$$\Rightarrow -1 + 1 = -B/A \Rightarrow \boxed{B = 0}$$

$$\textcircled{2} \quad \alpha\beta = c/a = C/A$$

$$-1 = C/A \Rightarrow \boxed{-A = C}$$

Q] What are the roots of the equation $4^x - 3(2^{(x+2)}) + 32 = 0$?

(A) 1,2

(B) 3,4

(C) 2,3

(D) 1,3

$$\begin{array}{c} 4^x - 3(2^{(x+2)}) + 32 = 0 \\ \downarrow \qquad \qquad \downarrow \\ (2^2)^x \qquad \qquad 2^x \cdot 2^2 \\ (2^x)^2 \end{array}$$

Soln : ① $2^x = t$

$$t^2 - 3[t \cdot 4] + 32 = 0$$

$$t^2 - 12t + 32 = 0$$

$$(t - 8)(t - 4) = 0$$

$$t = 4, 8$$

∴

$$t = 4$$

$$2^x = 2^2$$

$$\boxed{x = 2}$$

$$t = 8$$

$$2^x = 2^3$$

$$\boxed{x = 3}$$

Shrenik Jain — Study Simplified

Q] For the quadratic equation $4x^2 - 3x + 2 = 0$

whose roots are a and b. Find

1) $\frac{a}{b} + \frac{b}{a}$

2) $a^2b + b^2a$

(M1)

Roots $\rightarrow x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

(M2)

Sum of roots = $a + b = \frac{-(-3)}{4} = 3/4 //$

Product of roots = $ab = \frac{2}{4} = 1/2 //$

$$\textcircled{1} \quad \frac{a}{b} + \frac{b}{a} = \frac{a^2 + b^2}{ab} = \frac{(3/4)^2 - 2(1/4)}{1/2} = -7/8 //$$

$$\# \quad (a+b)^2 = a^2 + b^2 + 2ab$$

$$\rightarrow a^2 + b^2 = (a+b)^2 - 2ab$$

$$\textcircled{2} \quad a^2b + b^2a = ab[a+b] = \frac{1}{2} \left(\frac{3}{4} \right) = \frac{3}{8} //$$

Q] For what value of a will the quadratic equation $x^2 - 5x + a = 0$ intersect x-axis exactly once?

☒ (A) $a = 25/4$

(B) $a = 23/4$

(C) $a = 0$

(D) $a = -5$

$D = 0$

$b^2 - 4ac = 0$

$25 - 4(1)(a) = 0$

$a = 25/4$ ✓✓✓

Q] If the quadratic equation $px^2 + qx + r = 0$ has roots equal in magnitude but opposite in sign then

(A) $r=0$

~~(B)~~ $q=0$

(C) $p=0$

(D) N.O.T.A

①

$$\alpha = -\beta$$

②

$$\text{Sum of roots} = -q/p$$

$$\cancel{\alpha + \beta} = -q/p$$

$$0 = -q/p$$

$$q = 0$$

Cubic Eqⁿ

$$ax^3 + bx^2 + cx + d = 0$$

Roots: α, β, γ

$$\textcircled{1} \quad \alpha + \beta + \gamma = -b/a$$

$$\textcircled{2} \quad \alpha\beta\gamma = -d/a$$

$$\textcircled{3} \quad \alpha\beta + \beta\gamma + \gamma\alpha = c/a$$

Q] The condition for $x^3 - px^2 + qx - r = 0$

Having two of its roots equal to each other but of opposite sign is

(A) $r = pq$

~~(B)~~ $r^2 = 2p^3 + 2$

~~(C)~~ $r = p^2q$

~~(D)~~ $r = (pq)^2$

① $\alpha, \beta, \gamma \longrightarrow$ Roots

② $\boxed{\alpha = -\beta}$

③ $\Sigma \text{ of roots} = p/1$

$(\cancel{\alpha} + \cancel{\beta}) + \gamma = p$

$\boxed{\gamma = p}$

$$x^3 - px^2 + qx - r = 0$$

$$\rightarrow \boxed{y = p}$$

$$\rightarrow LHS = RHS$$

$$\cancel{p^3} - \cancel{p(p)^2} + q(p) - r = 0$$

$$qp - r = 0$$

$$\boxed{r = pq}$$

Inequality

$$\textcircled{1} \quad a > b \longrightarrow b < a$$

$$\textcircled{2} \quad a \neq b \longrightarrow a \leq b$$

$$\textcircled{3} \quad a \nless b \longrightarrow a \geq b$$

$$\textcircled{4} \quad a > b \text{ \& } b > c \longrightarrow a > c$$

$$\textcircled{5} \quad a < b \text{ \& } b < c \longrightarrow a < c$$

$$\textcircled{6} \quad a > b \longrightarrow \begin{array}{l} a + c > b + c \\ a - c > b - c \end{array}$$

$$(7) \quad a > b \text{ \& } c > 0 \rightarrow ac > bc$$

$$(8) \quad \boxed{a > b} \text{ \& } c < 0 \rightarrow ac < bc$$

$$(9) \quad a > b \text{ \& } c > d \rightarrow a + c > b + d$$

$$(10) \quad a > 0 \rightarrow -a < 0$$

$$(11) \quad a > b \rightarrow -a < -b$$

$$\boxed{a > b}$$

$$\textcircled{1} \quad \frac{1}{a} < \frac{1}{b}$$

$$\textcircled{2} \quad \frac{a}{c} > \frac{b}{c}, \quad \underline{c > 0}$$

$$\textcircled{3} \quad \frac{a}{c} < \frac{b}{c}, \quad c < 0$$

Q] If $(2x-3y < 7)$ and $(x+6y < 11)$ then which one of the following is correct?

(A) $x+y < 5$

☒ (B) $x+y < 6$

(C) $x+y \leq 5$

(D) $x+y \leq 6$

$$2x - 3y < 7$$

$$x + 6y < 11$$

$$3x + 3y < 18$$

$$x + y < 6$$

Q] $2(x+1) < 5x+3$

☒ (A) $x > -1/3$

(B) $x > 0$

(C) $x \geq -1/3$

(D) $x < -1/3$

$2x + 2$ < $5x + 3$

$-3x < 1$

$3x > -1$

$x > -1/3$

Q] Find the range of values of x that satisfy the on equations $-8 \leq 3x + 1 \leq 4$ and $-8 \leq 2x - 2 \leq 6$

(A) $2 \leq x \leq 3$

(B) $-2 \leq x \leq 4$

~~(C) $-3 \leq x \leq 4$~~

~~(D) $-3 \leq x \leq 1$~~

$$-8 \leq 3x + 1 \leq 4$$

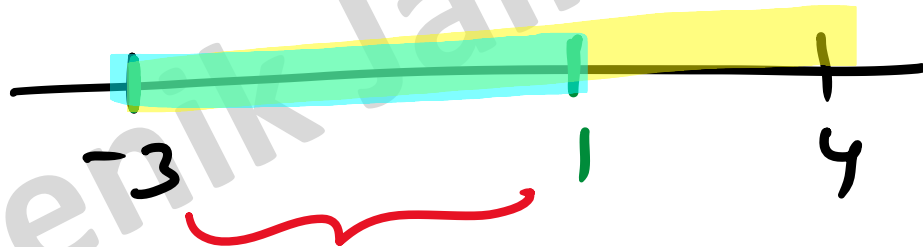
$$\Rightarrow \frac{-9}{3} \leq \frac{3x}{3} \leq \frac{3}{3}$$

$$\Rightarrow \boxed{-3 \leq x \leq 1}$$

$$\underline{-8 \leq 2x - 2 \leq 6}$$

$$\Rightarrow -6 \leq 2x \leq 8$$

$$\Rightarrow \boxed{-3 \leq x \leq 4}$$



Common range = ans $\Rightarrow -3 \leq x \leq 1$

Q] Find the range of x if $|x - 5| < 12$

(A) $-6 < x < 12$

(B) $-8 < x < 13$

~~(C) $-7 < x < 12$~~

(D) $0 < x < 12$

$x = 5 \rightarrow f = 0$

$x > 5$

$(x - 5)$

$x < 5$

$-(x - 5)$

$+(x - 5) < 12$

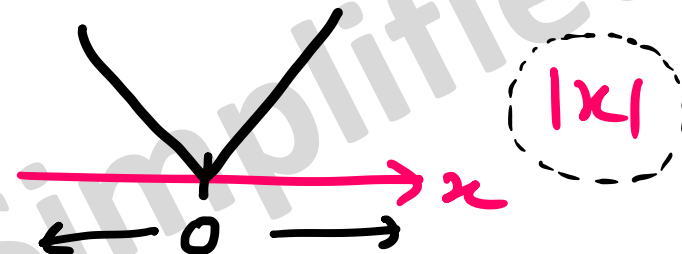
$\Rightarrow x < 17$



$-(x - 5) < 12$

$(x - 5) > -12$

$x > -7$



$|x| = \begin{cases} x, & x > 0 \\ -(x), & x < 0 \end{cases}$